

Technical Change, Job Tasks and Rising Educational Demands: Looking Outside the Wage Structure.[§]

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April 2005

Abstract

Empirical work has been limited in its ability to directly study whether skill requirements in the workplace have been rising, and whether these changes have been related to technological change. This paper answers these questions using a unique data set from West Germany that enables me to look at how skill requirements have changed *within* occupations and how these changes are related to workplace computerization. I show that occupations require more complex skills today than in 1979, and that the changes in skill requirements have been most pronounced in rapidly computerizing occupations. These patterns also exist *within* occupation-education groups and *within* occupation-age groups. The changes in occupational content account for about 50 percent of the recent educational upgrading in employment.

JEL-classification: J 24, J23, O30

Keywords: occupational skill requirements, skill-biased technological change

[§]**Acknowledgments:** The paper was written as part of the research project “ICT, Workplace Organization, and Wages Across and Within Skill Groups” commissioned by the Landesstiftung Baden-Württemberg foundation. I thank David Autor for his invaluable suggestions. I am also indebted to Daron Acemoglu, Miriam Beblo, Irene Bertschek, Sandra Black, Lex Borghans, Bernd Fitzenberger, Thomas Hempell, Francis Kramarz, Frank Levy, Elisabeth Mueller, Jörn-Steffen Pischke, Susanne Prantl, Bas ter Weel and Elke Wolf for their suggestions and comments.

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1 Introduction

In recent decades, the U.S. and U.K. have witnessed both a major increase in the supply of more educated workers and a rise in the return to education, supporting the argument that technological change has been skill-biased. However, although companies in continental European economies have had access to the same technologies as companies in the U.S. and the U.K., the wage structure in these countries has not changed accordingly - a puzzle that continues to motivate economic research.¹ But, to date, there has been no conclusive evidence on how skill requirements have changed in continental Europe and how these changes are related to technological changes.

In this study, I go beyond the analysis of wages and analyze directly changing occupational skill requirements using a unique data set from West Germany. The measures of skill requirements are based on the task-composition of occupations; that is, employees who participated in the survey indicated what they actually do in their jobs. Importantly, while wages typically incorporate non-skill characteristics such as the effects of unions and discrimination, the task-composition of occupations is unlikely to be influenced by such factors.

Another key advantage over the existing literature is the fact that I observe skill requirements *within* occupations. The occupational classifications are constant over time, so that detailed analyses of the changing skill requirement patterns within occupations can be carried out on the basis of the task descriptions. This property of the data set allows me to offer a detailed characterization of workplace changes, as well as their association with the introduction of computer technologies in the workplace. Owing to the task measure, I am able to show that the German labor market has witnessed changes in skill requirements similar to those in the U.S.

The analysis builds on the Autor-Levy-Murnane (2003) model, which proposes a nuanced version of the skill-biased technological change hypothesis. However, the evidence presented in this study improves considerably on the evidence offered by Autor, Levy and Murnane (2003), who infer the task content of individual occupations by linking CPS job titles to data from the Dictionary of Occupational Titles (DOT). This introduces considerable noise into the skill measurement for individual occupations. Importantly, it also precludes to a large extent a discussion on *within* occupational task changes. In contrast, the analysis in this paper is based on a consistent, well-defined, direct measure of occupational task inputs. Assuming that computer technologies substitute for certain tasks - those that are expressible in rules and thus programmable - and not for whole occupations, *within* occupational task changes are the natural dimension for analyzing questions of skill-biased technological change. Thus, by focusing the analysis on within-occupational task changes, this paper goes an important step further.

¹Comprehensive reviews of the skill-biased technological change literature can be found in Katz and Autor (1999) and Chennells and van Reenen (1999). Krueger and Pischke (1998), Card, Kramarz and Lemieux (1999), Acemoglu (2003), Piketty and Saez (2003), and Saez and Veall (2004), for example, discuss the puzzle of differential inequality trends in continental European countries and the U.S./U.K. in recent decades.

The results suggest that occupations involve greater complexity today than at the end of the 1970s. There has been a sharp increase in non-routine cognitive tasks such as doing research, planning or selling and a pronounced decline in manual and cognitive routine tasks such as double-entry bookkeeping and machine feeding. Importantly, most of the task changes have occurred within occupations and they have been most pronounced in occupations that have heavily introduced computer technologies, results consistent with the skill-biased technological change hypothesis. Neither changes in the education structure nor in the gender composition of occupations provide alternative explanations for the changing skill requirements. In contrast, the above patterns also exist *within* occupation-education groups and *within* occupation-age groups. Moreover, the analysis suggests that changing occupational skill requirements explain a significant part of the educational upgrading of recent decades. In addition, the changes in the occupational structure of employment between 1979 and 1998/99 suggest that there is evidence of “hollowing out” in the labor market.

The paper is organized in 6 sections. The next section discusses the related literature and introduces the “task framework” used in this study. Section 3 describes the data set and the variables. Section 4 presents stylized facts on occupational skill requirements, and educational and technological trends in West Germany since 1979. Section 5 investigates the relationship between computer use, occupational skill requirements and educational attainment on the basis of synthetic occupation groups. Section 6 presents the conclusions.

2 Occupational Skill Requirements and Rising Supply of Education

The effect of technological change on labor demand has always been a major concern of economic research. A central theme in this discussion is whether the restructuring and reorganization of workplaces owing to technological developments leads to skill upgrading or skill downgrading.² The discussion has intensified with the spread of computer technology in workplaces in recent decades. Based on the recent observed shifts in the earnings distribution in the U.S., non-neutral technological change, increasing the productivity of highly skilled employees more than that of less skilled workers, has been given particular attention. In addition, the “polarization” of the labor force has often been discussed.³

Empirical research has provided evidence of robust correlations between computer-based technologies and the use of highly skilled employees at various levels of aggregation, strengthening the hypothesis that recent technological change has been skill-biased. These studies emphasize the higher skills now required at the workplace owing to computeriza-

²A classical study is Braverman (1974), others are Spenner (1983) and Diprete (1988). Goldin and Katz (1996, 1998) provide a historical perspective on the relationship between technology and skill demand.

³See, for example, Levy and Murnane (1992).

tion.⁴ In addition, the stylized fact of rising returns to education in spite of an increasing supply of more educated workers supports the skill-biased technological change (SBTC) hypothesis.⁵ Katz and Autor (1999), Acemoglu (2002) and Chennells and van Reenen (1999) give comprehensive reviews of the literature on SBTC, covering the major studies in this field of research.

However, most studies in the SBTC literature use “traditional” skill measures to assess the skill level of employees, such as the proportion of production workers/non-production workers or blue-collar/white-collar workers.⁶ These classifications use divisions according to occupational groups which are of limited use in determining skill requirements. They document, for example, the structural shift toward increased employment of white-collar work in all major sectors of industrialized countries. Other studies rely on measures of formal education or wages.⁷ Education, however, is an input factor. Firstly, it is very likely that people with equal investment in their formal education attain different levels of skills. Each education group is therefore best characterized by a distribution of skills.⁸ In addition, skills that people bring to jobs in the sense of individual attributes - such as knowledge, abilities or capacities - do not necessarily coincide with the skills that are required to perform certain tasks at the workplace.⁹ Wages, on the other hand, may not reflect the “true” skill level of individuals either. This is not only the case in countries with centralized wage settings such as Germany and France. It may likewise be the case in

⁴The oppositional view has been taken by the over-education literature that states that skill requirements did not change considerably in recent decades and that the increased deployment of highly-educated employees resulted in them holding occupations that were previously performed by employees with lower education levels. Empirical studies are, for example, Rumberger (1987), Duncan and Hoffman (1981), Sicherman (1991), Alba-Ramirez (1993), Verdugo and Verdugo (1989) and Groot and Maassen van den Brink (2000). The empirical studies have been criticized for the way they operationalize over-education (see, among others, Halaby, 1994, Smith, 1986). In addition, results of recent studies that take unobserved heterogeneity into account or use instrumental variable techniques question the positive wage effects of over-education found in cross-section analyses (see e.g. Bauer, 2002). These results suggest that part of what is referred to over-education simply reflects the heterogeneity of individual abilities and skills within particular educational qualifications.

⁵The analysis in the present study focuses on technological change (as opposed to de-industrialization and globalization as alternative explanations discussed in literature for the skill upgrading in recent decades), measured by workplace computerization, because it is the only explanation that generates predictions about within-occupational task changes. De-industrialization and globalization, on the other hand, emphasize between-industry or between-occupation developments.

⁶For example Berman, Bound and Griliches (1994), Berndt, Morrison and Rosenblum (1994) and Berman, Bound and Machin (1998).

⁷For example, Autor, Katz and Krueger (1998).

⁸The evidence presented by Katz and Murphy (1992), Levy and Murnane (1992) and Juhn, Murphy and Pierce (1993) points to the importance of distinguishing between formal education and skills in the context of wage inequality. Murnane, Willett and Levy (1995) present evidence of the growing importance of cognitive skills. The diversity of skills within demographic categories is also emphasized by Heckman and Sedlacek (1985).

⁹See Spenner (1983, 1990).

other countries owing, for example, to discrimination or segregation in the labor market.

The recent study by Autor, Levy and Murnane (2003), however, now offers a framework that makes it possible to analyze occupational skill requirements directly.¹⁰ The major feature of their framework is that they conceptualize work as a series of tasks; with this, one can analyze the recent changes in the task composition of occupations directly.

Building on this model, I use direct measures of occupational skill requirements that are based on the activities people perform on the job. These activities are classified in five skill categories: *non-routine analytical* tasks such as research, planning or evaluation activities; *non-routine interactive* tasks such as selling, the coordination and delegation of work; *routine cognitive* tasks such as double-entry bookkeeping and calculating; *routine manual* tasks such as machine feeding or running a machine and *non-routine manual* tasks such as housekeeping or restoring houses.

The terms *routine* and *non-routine* characterize the relationship between the respective task measure and IT. Both manual and cognitive routine tasks are well-defined in the sense that they are expressible in rules, which makes them easily programmable, therefore computers can perform them at economically feasible costs (Levy and Murnane, 1996). Hence, routine tasks are subject to substitution by computer capital. *Non-routine* tasks are not well-defined and programmable. Therefore, at present, they cannot be accomplished by computers. However, computer capital is complementary to non-routine cognitive tasks, both analytical and interactive, in the sense that computer technology increases the productivity of employees performing those tasks. The term *analytical* refers to the ability of workers to think, reason and solve problems encountered at the workplace. The term *interactive* refers not only to communication skills - that is, the ability to communicate effectively with others through speech and writing - but also to the ability to work with others, including co-workers and customers.¹¹

The scope for substitution is thus limited to certain tasks. This *limited substitution* relationship (Bresnahan, 1999) between IT and occupational tasks shifts the demand for labor toward employees with higher levels of educational attainment who are presumed to have a comparative advantage in performing non-routine cognitive tasks. Autor, Levy and Murnane (2003) present a general equilibrium model that is the foundation of this informal reasoning showing how computerization (owing to the exogenous declining price of computer capital) alters the allocation of labor across different task inputs.

Wage trends in Germany are often considered different from developments in other countries, even though Fitzenberger (1999) and Fitzenberger et al. (2001) provide evidence

¹⁰This framework sheds light on the “black box” that typically encloses studies on SBTC, as was, for example, expressed by Bresnahan (1999, p. 340): “...(skill-biased technological change) also tends to be something of a residual concept, whose operational meaning is often *labor demand shift with invisible cause*”.

¹¹Case studies identified analytical and interactive skills as the “key” skills required by modern workplaces in industrialized countries (for example, Hirschhorn, 1984, Stasz, 1997, 2001).

that there have been changes in the wage structure in West Germany in recent decades.¹² However, the changes have been small by international standards (see, for example, Prasad, 2004, and Abraham and Houseman, 1995). Wages of employees with a medium level of education deteriorated after 1980 relative to both employees with high and low levels of education, and the relative wage position of the bottom part of the wage distribution of employees with high levels of education has slightly deteriorated over time. In addition, the wage dispersion among medium- and high-educated employees increased over time. The main difference between West Germany and other countries is the experience of employees with low levels of education. In contrast to the wage decreases in most other countries, their wages slightly increased and the wage dispersion within this group of workers remained stable over time. However, this group of workers experienced a sharp increase in unemployment since the 1980s.¹³ The lack of adjustment of wages of employees with low levels of education is often explained by union wages that are binding floors for low-wage earners.¹⁴ Others proposed “social norms” as alternative explanations for the divergent developments in the wage structure in the U.S. and Germany (or continental European countries in general).¹⁵

An important advantage of the skill measure in this study is that centralized wage setting institutions are unlikely to influence the task-composition of occupations. Unions bargain over wages but they do not have influence on decisions concerning, for example, technology diffusion within companies. Works councils, which are mandated in Germany for companies with more than 5 employees, typically have an influence on questions of technological changes within companies.¹⁶ However, recent studies investigating the relationship between the existence of works councils and firm productivity find positive or insignificant effects.¹⁷ Overall, these results suggest that works councils do, on average, not hamper productivity enhancing measures in companies. In addition, Bemmels and Reshef (1991) find that an effective means of avoiding employee resistance to technological changes is to involve them in the decision making process; employees tend even to react

¹²The most comprehensive analyses of wage trends exists for the U.S., for example by Bound and Johnson (1992), Katz and Murphy (1992), Levy and Murnane (1992) and Juhn, Murphy and Pierce (1993). Gottschalk and Smeeding (1997) provide an international comparison of earnings and income developments since the early 1980s.

¹³The European unemployment problem and the U.S. inequality problem are often referred to as two sides of the same coin (Krugman, 1994, Freeman, 1995). Criticism of this hypothesis can be found, for example, in Nickell and Bell (1996) and Krueger and Pischke (1997). Card, Kramarz and Lemieux (1999) provide evidence for France (often considered to be similar to West Germany with respect to its labor market institutions) that is inconsistent with the “two-sides-of-the-same-coin” hypothesis.

¹⁴Empirical studies by Falk (2001), Falk and Koebel (2001), Fitzenberger (1999) and Kaiser (2000) investigate SBTC in West Germany. The picture they present is consistent with the view that recent technological change in West Germany has also been skill-biased.

¹⁵See, for example, Piketty and Saez (2003), Saez and Veall (2004) and Krueger and Pischke (1998).

¹⁶See, for example, Freeman and Lazear (1995).

¹⁷See Addison, Schank, Schnabel and Wagner (2003) for a comprehensive review of this body of the literature.

positively to technological changes if these are associated with an increase in workers' skill requirements.

3 Data Set and Definition of Variables

The analysis is based on the "Qualification and Career Survey" which is a survey of employees carried out by the German Federal Institute for Vocational Training ("Bundesinstitut für Berufsbildung, BIBB") and the Research Institute of the Federal Employment Service ("Institut für Arbeitsmarkt- und Berufsforschung, IAB"). It includes four cross-sections launched in 1979, 1985/86, 1991/92 and 1998/99 each covering about 30,000 individuals (men and women).¹⁸

Employees are classified according to their occupation; the classification of occupational titles corresponds to that of the German Federal Employment Bureau, 1988. In the present study the 2-digit level of classification is used which includes around 80 different occupations.¹⁹ The micro level data of the 4 cross-sections are aggregated into occupation cells and the group means are used for the analysis. The data cover a wide range of industries, both services and manufacturing, listed in Table A in the appendix.

The data set is particularly suited to analyze changes in skill requirements within occupational since occupations in all four waves are categorized according to the 1988 classification. The constant occupational titles thus provide the reference point for the analysis. A major improvement over previous work is that survey respondents indicated what kind of activities they perform on the job. It is very unlikely that this causes an underestimation of true changes in job content as in the Dictionary of Occupational Titles (DOT), the description of occupations often used by researchers in the U.S. for questions related to skills.²⁰ The credibility of the analysis in the present study would be impaired if the answers of survey participants with high levels of education were systematically biased toward analytical and interactive activities. This is unlikely as survey participants only indicate whether they perform certain activities or not and do not assign scores to the different measures. In addition, most of the analysis is performed in "first-differences".

¹⁸The target population is not uniform within the four waves. Due to this changing sample design the sample used in the present study had to be restricted to West-German residents with German nationality, in other words East-German residents and non-German employees are excluded from the sample since these groups of employees were not interviewed in every wave. Moreover, the sample does not include self-employed and unemployed persons, employees with agricultural occupations and employees working in the agricultural sector. In addition, persons younger than 18 and older than 65 are excluded from the sample.

¹⁹I use the 2-digit level of occupational classification rather than the 3rd or 4th level in order to minimize problems of measurement error.

²⁰In the DOT, experts assign scores to different indicators characterizing the occupations. It is well known that this process encourages analysts to underestimate the true changes in job content. Moreover, occupational titles in the DOT are not consistent over time (for detailed criticism see Spenner, 1983, and references cited there).

The reporting bias therefore would only pose a problem if it changed over time. As will be presented later on, the empirical results even hold within-occupation-education groups.

The most important variables for the present analysis are the measures of occupational skill requirements, the measure of technology, and the level of employees' formal education.

Occupational Skill Requirements: Occupational skill requirements are measured by the workers' job duties, depicted in the survey by the activities that employees have to perform at the workplace.²¹ I pool these activities into five task categories. These are: *analytical* tasks, *interactive* tasks, *routine cognitive* tasks, *routine manual* tasks and *non-routine manual* tasks. Table 1 illustrates the assignment of activities to the five categories. On the individual level i , the task measures ($Task_{ijt}$) are defined as:

$$Task_{ijt} = \frac{\text{number of activities in category } j \text{ performed by } i \text{ in cross-section } t}{\text{total number of activities in category } j \text{ at time } t} * 100 \quad (1)$$

where

$t = 1979, 1984/85, 1991/92$ and $1998/99$, and

$$j = \begin{cases} 1 & : \text{ non-routine analytic tasks} \\ 2 & : \text{ non-routine interactive tasks} \\ 3 & : \text{ routine cognitive tasks} \\ 4 & : \text{ routine manual tasks} \\ 5 & : \text{ non-routine manual tasks.} \end{cases}$$

For example, if the analytical task category includes 4 activities and employee i indicates that she performs 2 of them, her analytical task measure is 50.

The data set does not include information about the time spent on different activities. In addition, while most questions remained the same over time, there were some changes in questions concerning the activities employees perform at the workplace. In order to be consistent, I reduced the activities in each category j to those that are comparable over time.

Computer use: The data set includes detailed information on the tools and machines used by the employees at the workplace. The focus in the present study is on the use of computers, terminals and electronic data processing machines. Based on these variables a dummy variable for computer use is generated indicating whether or not the employee uses one of the above devices on-the-job.

²¹I use the terms "skill requirements" and "skill/task inputs" interchangeably throughout the paper, although the term "skill/task inputs" is strictly speaking only correct. In order to being able to speak of "skill requirements", I would need information about task prices.

Formal Educational Attainment: The data set contains detailed information on the vocational attainment of employees. The employees are classified into three qualification groups according to their vocational education (school qualifications are not considered):²²

- People with low levels of education, that is, people with no occupational training.
- People with medium level of education, that is, people with a vocational qualification who might have either completed an apprenticeship or graduated from a vocational college.
- People with high level of education, that is, people holding a degree from a university or technical college.

These variables are dummy variables, taking on the value 1 if the employee falls within the particular education level.²³

4 Overall Trends in Educational Supply, Occupational Skill Requirements and the Evolution of IT at the Workplace

As in most industrialized countries, in West Germany the labor force has witnessed a sizeable relative increase in the proportion of workers with high levels of education (see Table 2, Panel A).²⁴ The proportion of the workforce holding a university degree or a qualification from a technical college increased from about 8 percent in 1979 to more than 16 percent in 1999, whereas the proportion of employees without formal educational attainment experienced a substantial decline. However, workers with a medium level of education who either completed an apprenticeship or have a qualification from a vocational college still represent the largest proportion of the workforce.²⁵

²²Most studies on West Germany use this classification rather than years of schooling because it is more appropriate to the system of vocational training. See Card (1999) for a detailed discussion of different measures of education in different institutional settings.

²³Theoretically there should be 332 observations in the stacked data set (83 occupations observed in four waves). It turns out that 94.5 percent of the occupations are observed in all waves, whereas 1.5 percent of the occupations are observed only once, 1.2 percent twice and 2.8 percent are observed three times. One might wonder whether there are occupations that disappear over time and others that are newly created, in particular, as there were a number of “information technology” occupations created by the German Federal Institute for Vocational Training in recent years. The disappearance of one occupation is most arguably attributable to structural change, namely the occupation in which workers prepared minerals (“Mineralaufbereiter”). None of the occupations that appeared over time was one of the newly-founded “information technology” occupations. Overall, the occupations that were observed less than 4 times seem to be a random draw. In particular, the pattern of their appearance is clearly not driven by the question of interest.

²⁴The descriptive evidence is based on the individual level data.

²⁵Spitz (2003) provides more descriptive details based on this data set. The descriptive findings are comparable to developments in other industrialized countries. For example, West Germany witnessed a

While there was this educational upgrading of the labor force, there was also a considerable change in aggregate skill requirements. Table 3, Panel A, shows the trends in aggregate skill inputs. The analytical task measure grew on average by 0.5 percentage points between 1979 and 1999, and the interactive task measure by 1.3 percentage points. In contrast, the requirements for routine cognitive and routine manual skills have decreased during that period with an average annual decline of 0.7 percentage points each. The trend in the requirements for non-routine manual skills is less clear. The overall period, however, suggests an increase of around 0.6 percentage points annually.

One important source of variation in workplace tasks is the substantial increase in interactive tasks, particularly in the 1990s. A detailed inspection of this category shows that tasks such as advising customers, organizing or coordinating increased in importance in the 1990s. By contrast, the increase in interactive task measure between 1985/86 and 1991/92 was mainly due to more people being involved in selling and in negotiating with others. The main source of the increase in the analytical task measure was a rise in employees doing analyses and research, whereas activities such as using and interpreting rules remained relatively stable over time.

Table 3 also shows the aggregate trends in skill inputs for each education group separately (Panel B-D). For each education group the analytic and interactive task measure increased over time, whereas the cognitive and manual routine task inputs declined. Figures 1-4 illustrate this development by showing the absolute changes in aggregate skill inputs between 1979 and 1998/99. Figure 1 shows the overall trends, and Figures 2-4 show the development for each education group. One difference between groups is that the decline of routine cognitive task inputs and the increase in interactive and analytical task inputs was more pronounced for employees with high levels of education than for the other two groups. Employees with medium levels of education witnessed the greatest decline in routine manual activities.

In contrast with a supply story, this overall pattern does not suggest that high-educated employees perform more of the tasks that used to be done by medium-educated employees at the end-1990s. In contrast, the more pronounced development toward analytic and interactive activities and away from routine cognitive activities suggests that overall skill requirements were rising faster for high-educated employees than for the other groups.

The argument could be made that these overall developments reflect cohort effects, that is, unobserved heterogeneity owing to, say, younger entry cohorts having better educational opportunities and therefore higher levels of analytical activities. To evaluate this possibility, Table 4 shows the trends in occupational task inputs for cohorts, defined by year of birth. The first birth cohort are individuals born before 1940, the second those born between 1940 and 1949, the third those born between 1950 and 1955 and so on.

substantial increase in white-collar occupations and a corresponding reduction in blue-collar occupations in the last two decades of the twentieth century. The occupational group of professionals, technical workers, managers and administrators saw the highest increase as a fraction of the workforce. This group places the highest emphasis on employees with high levels of education. Formal degrees seem to have become more important in all occupational groups. However, the descriptive figures also demonstrate that each occupational category includes employees with all levels of formal education.

One follows one birth cohort over time by moving horizontally within the same row. One follows the same age group by moving diagonally upwards (employees born between 1956 and 1961 were between 18 and 23 years old in 1979, employees born between 1962 and 1968 were between 18 and 23 years old in 1985/86).²⁶

As the figures in Table 4 show, the trend toward analytical and interactive task inputs, and away from cognitive and manual routine activities occurred both within cohorts and within age groups. The overall trends are therefore not only a reflection of cohort effects. Older cohorts experienced the same trends. For analytical and interactive as well as routine manual task inputs, changes within cohorts were even more pronounced than changes within age levels. Within cohorts, analytical task inputs, for example, increased by around 0.6 percentage points annually on average, whereas the annual increase within age groups was 0.5 percentage points on average. These results suggest that time effects appear to be more important than age or cohort effects *per se* in explaining changes in task inputs.

These trends in aggregate skill requirements may result from transformations along two margins: First, changes in the occupational structure of employment, and second, changes in skill requirements within occupations.²⁷ As the results of the shift-share analysis in Table 5 show, most of the aggregate changes in skill requirements result from within occupational changes in task measures.²⁸ The last row, which shows the results for the entire period, clearly illustrates this point. For example, between occupational shifts represent around 15 percent of aggregate changes in the analytical task measure whereas within occupational task changes account for around 85 percent. In the case of changes in interactive (routine manual) skill requirements the values are quite similar with around 13 (14) percent attributable to the between shift and around 87 (86) percent to the within shift. The results for the routine cognitive task measure are even more pronounced,

²⁶Within cohorts, changes in task inputs are attributable to age and time effects. The age effect describes how the task inputs of a given cohort changes as the cohort ages. The time effect describes how task inputs for a given cohort shift due to, for example, macroeconomic shocks. Changes in task inputs within an age group, on the other hand, are due to cohort or time effects. Cohort effects describe differences between cohorts that may, for example, be due to changes in educational opportunities. It is well known that the three components - time, cohort and age effect - are not separately identifiable without additional prior assumptions. This results from the identity that links birth year c , age a and calendar year t : $t = c + a$.

²⁷Autor, Levy and Murnane (2003) provide a comprehensive analysis of the first source of variation for measuring changes in aggregate skill requirements, that is changes in the occupational structure of employees, but are unable to examine the second in detail.

²⁸The shift-share analysis decomposes the change in aggregate use of task j between time t and $t-1$, $\Delta T_{jt} = T_{jt} - T_{jt-1}$, into a term reflecting the reallocation of employees between occupations and a term reflecting changes in task j within occupation. The mathematical formulation is: $\Delta T_{jt} = \sum_c (\Delta E_{ct} \bar{\gamma}_{cj}) + \sum_c (\Delta \gamma_{cjt} \bar{E}_c) = \Delta T_{jt}^b + \Delta T_{jt}^w$, where c indexes occupations, E denotes employment and E_{ct} is the share of employment in occupation c in total employment at time t , γ_{cjt} is measure of task j in occupation c at time t . An overstrike denotes an average over time, that is $\bar{\gamma}_{cj} = (\gamma_{cjt} - \gamma_{cjt-1})/2$ and $\bar{E}_c = (E_{ct} - E_{ct-1})/2$. ΔT_{jt}^b reflects the change in aggregate employment of task j attributable to changes in the occupational distribution of employment and ΔT_{jt}^w reflects the within-occupation task changes.

indicating that between-occupation shifts account for less than one percent of aggregate changes in routine cognitive skill requirements. For this task measure, it is also informative to have a look at the two subperiods 1985/86-1991/92 and 1991/92-1998/99 because between occupational results there point to slight increases in the requirements for routine cognitive skills, a pattern that has been counteracted by the within occupational task shifts. The overall result of this table of predominantly within occupational task shifts is largely in favor of technological developments rather than changes in final demand as the potential cause for changing skill requirements.

One important argument in this study is that the changes in the task composition of occupations toward analytical and interactive activities induced labor demand shifts toward employees with high levels of education who are viewed as having comparative advantages in performing non-routine cognitive tasks. Table 3, Panel B-D, shows that the analytical and interactive task inputs are the highest for employees with high levels of education in each wave. The descriptive evidence thus confirms the view that the higher the educational attainment, the higher the measures in analytical and interactive tasks. In contrast, the figures indicate that employees with low levels of education are mainly occupied with routine cognitive, routine manual and non-routine manual tasks. Employees with medium levels of education have relatively high measures for all five task categories, but most interestingly, the value of their task measures for manual and cognitive routine activities are even higher than those for employees with low levels of education.

The educational upgrading and the changes in occupational skill requirements took place at the same time as information technology began to spread at the workplace. Whereas at the beginning of the IT revolution mainframe computers dominated the data-processing units of large firms, personal computers began to spread to business users from the late 1970s onwards. Owing to the steady fall in prices, this spread has become more pronounced. Table 2, Panel B, shows the percentage of computer users at work. The table shows that within twenty years, more than half of the workforce has come to use computers at work.²⁹ Between 1979 and 1999, the spread of computers increased on average by more than 40 percent per annum. Table 2, Panel B, also shows that the level of computer use increased with the educational attainment of employees. In 1979, more than 10 percent of employees with high levels of education already used a computer at the workplace compared to less than 4 percent of employees with low levels of education and around 6 percent of employees with medium levels of education. This proportion had increased to more than 80 (30, 55) percent of employees with high (low, medium) levels of education in 1999. However, it is worth noting that, with an increase of around 45 percent per annum, the pace of computer diffusion was most pronounced among employees with low levels of education between 1979 and 1999 (compared to around 42 percent per annum for employees with medium levels of education and 30 percent for employees with high

²⁹The figures on computer penetration at the workplace are similar to those reported for other countries, for example, in Card, Kramarz and Lemieux (1999).

levels of education).³⁰

The descriptive evidence shows that educational upgrading, increased demand for analytical and interactive activities and the spread of computer technologies evolved together in recent decades. This is consistent with the argument that IT increases the demand for employees with high levels of education through shifting the task composition toward analytical and interactive activities.

5 Skill Requirements, Education and Technology in the Workplace

5.1 Technological Change and Changes in Occupational Skill Requirements

Based on the model by Autor, Levy and Murnane (2003) outlined in Section 2, there are two empirically testable hypotheses: (1) that IT substitutes for routine manual and routine cognitive activities, and (2) that IT is complementary to analytic and interactive activities. The framework emphasizes that the causal force by which IT affects skill demand is the declining price of IT. As the price of IT falls steadily, these two mechanisms have raised relative demand for employees with high levels of education who are assumed to have a comparative advantage in performing analytical and interactive activities.

The analysis that follows will investigate this substitution and complementarity hypothesis. Because there are no hypotheses on the relationship between IT and changes in non-routine manual task inputs derived from this theoretical model, I do not analyze this relationship in detail. However, as the relationship might still be interesting, I will come back to this issue in Footnote 35 at the end of this section.

Table 6, Panel A, shows the first-difference relationship between workplace computerization and changes in occupational skill requirements. Each column represents a separate OLS regression of the annual changes in occupational task measures on the annual changes in occupational computer use. Annual changes are estimated between successive waves, that is between 1979 and 1985/86, between 1985/86 and 1991/92 as well as between 1991/92 and 1998/99. The regressions are based on the stacked data set. They are performed including time dummies for 1985/86-1991/92 and 1991/92-1998/99 capturing the trend in within occupational tasks changes for the corresponding time period relative to the base period 1979-1985/86.

The results show that occupations that saw greater increases in computerization witnessed significantly larger increases in analytical and interactive task requirements, whereas they

³⁰Unreported results show that computer adoption has been quite different for different occupational groups. It has been particularly pronounced in professional, technical, managerial, administrative and clerical occupations. The spread of computer capital has been much broader among clerical occupations than among sales occupations. In contrast, to the wide spread of computers in most white-collar bureaucracies, operatives and crafts people witnessed a much slower penetration rate of IT at the workplace.

witnessed greater declines in routine manual and routine cognitive task requirements. The coefficients are not only statistically significant but also economically large. For example, they indicate that 50 percent of changes in analytical task inputs were accounted for by computerization.³¹ Similar calculations show that 47 percent of changes in interactive task inputs are accounted for by computerization and it explains 90 percent of the decline in routine cognitive skill requirements. In the case of routine manual task inputs, computerization more than fully accounts for the observed task changes.

The time dummies show that, conditional on workplace computerization, the trend change in analytical skill requirements was negative in both periods 1985/86-1991/92 and 1991/92-1998/99 relative to the base period of 1979-1985/86 indicating that computerization more than fully accounts for the trend toward analytical task inputs in these later periods. In contrast, the trend change toward interactive skill inputs accelerated with time even after conditioning on computerization. In the routine task equations, the coefficients of the time dummies are mostly insignificant, except for a large positive trend in 1991/92-1998/99 in routine cognitive skills.

Panel B shows the result when the lagged annual changes in computer use within occupation cell is used as regressor instead of contemporaneous changes in workplace computerization. The results of the lagged specification confirm the previous findings reported in Panel A, with the exception of the analytical task equation that now has an insignificant coefficient. The coefficient in the routine cognitive equation even increases in absolute value suggesting that there might be a time lag until the full impact of computerization is reflected in occupational skill requirements.³² These lagged results favor the argument derived from the theory that workplace computerization, brought about by the declining prices of IT-equipment, induced task shifts and not vice versa.

One might wonder whether changes in occupational skill requirements are implicitly captured by changes in the educational structure or changes in the gender structure of occupations. Table 7, Panel A, shows the result of a richer specification that includes the respective controls. Neither of the additional regressors alters the qualitative relationship between computerization and changes in occupational skill requirements found in the bivariate regressions, and even the size of the coefficients are relatively insensitive to the additional controls. Only in the analytical task equation, does the coefficient drop by a bit more than ten percent. Changes in the proportion of employees with high levels of education turn out to play a significant role with respect to changes in interactive and analytical skill requirements, whereas changes in the proportion of employees with medium levels of education appear to be bad predictors of changes in occupational task inputs.

³¹At the bottom of Table 6, the unconditional (weighted) means of the dependent variables are shown. The figures indicate an average annual increase in the analytical task measure of 0.425 percentage points. Using the coefficient of 0.086 and the mean value of changes of computer utilization of 2.465 percentage points, this implies that around 50 percent of changes in analytical tasks is accounted for by changes in IT usage.

³²Unreported regressions in which the lagged proportion of computer users are included as an alternative measure confirm these results.

Changes in the proportion of female employees are even negatively related to changes in analytical skill requirements, hence they fail to provide an alternative explanation for increasing analytical skill requirements. However, occupations with greater increases in the proportion of female employees witnessed relatively larger declines in routine cognitive skill requirements.³³

Panel B shows the results when lagged annual changes are included as regressors in the analysis instead of contemporaneous changes. The results for the relationship between computerization and changes in occupational skill requirements are similar to those reported in Table 6, Panel B. For changes in analytical skill requirements, contemporaneous changes in all variables seem to pick up more of the relevant information. Unlike the variable for workplace computerization, a comparison of the results for education and gender variables with those in Panel A reveals, however, that they are not insensitive to the change in specification.

Table 8 shows the relationship between workplace computerization and changes in occupational skill requirements for each education group separately. For each occupation-education group, the results are consistent with the hypotheses of a complementary relationship between computer technology and analytical and interactive activities, while computer technology and manual and cognitive routine tasks appear to be substitutes. However, for part of the coefficients, the level of significance drops considerably. This is particularly pronounced for the group of employees with a high level of education.³⁴ I see two potential explanations for this finding. First, that changes in the task composition owing to workplace computerization are less pronounced for employees with high levels of education because their values of the task measures were already at the extreme of the distribution in 1979. Second, that for this group of workers the analysis might be particularly impaired by the fact that the task measures do not include information about the time spent with the different activities. This time dimension might be more important for employees who always had a high number of activities that they perform. In line with these arguments, the results for employees with a low level of education are the clearest with respect to the magnitude and significance of the coefficients. For this group of employees, workplace computerization has had a large effect on the occupational production function.

I also analyzed the relationship between workplace computerization and changes in occupational skill requirements for each birth cohort introduced in Table 4. The (unreported) results continue to suggest a complementary relationship between computer technologies and analytical and interactive activities and a substitutive relationship between computer technologies and cognitive and manual routine activities within each occupation-cohort

³³I also performed regressions that include changes in work-based learning (average years of work experience and average years of tenure with current employer) into the analysis. Both measures of work-based learning were not significantly related to changes in occupational task inputs. The results are not reported because the inclusion of these additional variables did not alter the main findings reported in Table 7.

³⁴Autor, Levy and Murnane (2003) report similar findings on the industry level.

group. Therefore, the previous findings do not seem to be a reflection of cohort effects.³⁵

5.2 Contribution of Changes in Occupational Skill Requirements to the Educational Upgrading

In this section, I calculate the potential contribution of shifts in occupation skill requirements to shifts in employment of labor with high levels of education and with medium levels of education. This is done by firstly estimating an equation of educational requirements as a function of task inputs, an exercise that aims to translate occupational skill requirements into education-equivalents. The regression equation is:

$$ED_{ict} = \alpha_{0it} + \sum_{j=1}^4 S_{jct} \alpha_{ijt} + v_{ict} \quad (2)$$

$t = 1979, 1985/86, 1991/92, 1998/99$

$j = 1, \dots, 4.$

$i = \text{high or medium}$

$c = \text{occupation}$

$$j = \begin{cases} 1 & : \text{non-routine analytic task measure} \\ 2 & : \text{non-routine interactive task measure} \\ 3 & : \text{routine cognitive task measure} \\ 4 & : \text{routine manual task measure} \end{cases}$$

where ED_{ict} is the proportion of employees with education level i in occupation c at time t . S_{jct} is the measure of task j in occupation c at time t . The α_{ijt} s estimated in the regression provide an estimate of the demand for employees with high (medium) level of education as a function of occupation skill requirements. This is what Autor, Levy and Murnane (2003) call a “fixed coefficient” model because, by assuming the α_{ijt} s to be constant over the period, it neglects the impact of task prices on task demand. If

³⁵Up to now, I completely neglected the relationship between non-routine manual tasks and the introduction of computer technologies in the workplace because the theoretical model does not make predictions about this relationship. However, it might still be interesting to analyze, even without prior hypothesis. The result of a regression similar to those reported in Table 6, Panel A, shows a positive but insignificant (coefficient: 0.126; standard error: 0.086) relationship between workplace computerization and changes in non-routine manual tasks. This result is robust to the inclusion of additional controls (analogously to the regressions presented in Table 7, Panel A) and for each education group separately (analogously to Table 8).

the prices of analytical and interactive task inputs relative to routine task inputs has risen, for example, this model will lead to an underestimation of shifts toward analytical and interactive tasks and thus to an underestimation of the educational upgrading of the workforce.

The estimated coefficients α_{ijt} of equation (2) are used to predict changes in the demand for employees with high and medium levels of education.

$$\Delta \hat{ED}_{ic\tau} = \sum_{j=1}^4 \Delta S_{jc\tau} \hat{\alpha}_{ij(t-1)} \quad (3)$$

$\tau = t - (t - 1)$ with $t = 1979, 1985/86, 1991/92, 1998/99$.

The results of estimating equation 2 separately for employees with high and medium levels of education and for each wave are shown in Table B in the appendix. Panel A presents the results for the proportion of employees with high levels of education. In each wave, the results indicate a strong (and mostly highly significant) positive relationship between analytical and interactive task inputs and the proportion of employees with high education levels. In 1979, for example, a one percentage point increase in analytical skill requirements results in a 1.3 percentage point increase in the demand for employees with high level of education. The results for the routine cognitive and routine manual task measures are also in line with the *a priori* expectations, although the coefficients are often insignificant.

Panel B shows the results for the proportion of employees with medium level of education. With most of the coefficients being insignificant, the results are less clear-cut than those for employees with high levels of education. The overall picture suggests, however, a positive relationship between routine cognitive and routine manual task inputs and the proportion of employees with medium levels of education, whereas the relationship with respect to analytical and interactive task inputs is negative.

Panel A of Table 9 shows the observed annual changes in the proportion of employees with high and medium level of education for the subperiods 1979-85/86, 1985/86-91/92, 1991/92-98/99 and for 1979-1998/99. The pace of educational upgrading was relatively stable since the mid-1980s with average annual changes of 0.5 percentage points. Between 1979 and 1998/99 the proportion of employees with high level of education grew on average by 0.4 percentage points per annum. The proportion of employees with medium levels of education decreased considerably between 1979 and 1985/86, increased between 1985/86 and 1991/92 and declined again in the 1990s. Over the whole period, however, this resulted in an average annual decrease of 0.1 percentage points.

In what follows, I will largely concentrate the discussion of results on the whole period 1979-1998/99, although the results for the different subperiods are mostly comparable. However, macroeconomic shocks might affect the outcomes in these subperiods such as the recessions that West Germany experienced at the beginning of the 1980s and around

1992/1993, and also the unification boom just at the beginning of the 1990s might be important with this respect.

Two different measures of $\Delta S_{j\epsilon\tau}$ are used to calculate $\Delta \hat{E}D_{ic\tau}$ of equation (3): first, observed changes in occupation skill requirements and second, predicted changes in occupation skill requirements implied by computerization. In the calculations that follow, I will only use the α_{ijs} of the year 1979. The relationship between task inputs and education in 1979 is closest to the pre-computer era. Desktop computing, for example, only became widespread in the 1980s and 1990s.

Panel B shows the observed changes in occupation skill requirements. Requirements for analytical and interactive skills within occupation grew between 1979 and 1998/99, with the pace of increases in interactive inputs accelerating steadily. The figures show a steady decline in occupation requirements for routine cognitive skills and also for routine manual skills.

Inserting the observed changes in occupation skill requirements, together with the $\hat{\alpha}$ s of Table B (appendix-first column), into equation (3) shows that the observed changes in occupational skill requirements account for nearly 50 percent of the changes in the proportion of employees with high level of education between 1979 and 1998/99 (Panel C). With respect to the observed changes in the proportion of employees with medium levels of education, observed changes in the occupational skill requirements even accounted for around 65 percent of observed changes. However, one has to keep in mind that the overall employment changes for this group of employees has been relatively small.

Panel D shows the results of the predicted changes in occupation skill requirements that are implied by computerization. These figures are based on the regression specification shown in Table 6, Panel A. The overall picture complies with the figures in Table 9, Panel B. Computerization implies an increase in occupational requirements for analytical and interactive tasks and a reduction in routine manual and routine cognitive tasks. The deviation between observed changes in skill requirements and the predictions implied by computerization is most pronounced in the nineties, where the predicted figures show a shift toward routine cognitive tasks.

Panel E shows the proportion of observed changes in the share of employees with high and medium level of education explained by predicted changes in occupation skill requirements implied by computerization. Between 1979 and 1998/99, occupational task changes induced by computerization account for more than 40 percent of the overall skill upgrading. This figures reaches with 47 (32) percent its height (low) in the first and last (second) subperiod. With respect to changes in the proportion of employees with medium levels of education, task shifts induced by computerization account for nearly 50 percent of the observed pattern of employment.

Given that these changes in skill requirements are only those within occupations, these figures show the substantial economic impact of changes in the skill requirements on the educational upgrading of the workforce. In addition, the analysis shows the role of workplace computerization in reshaping the occupational production process. Changes in

occupational skill requirements implied by computerization account for 85 percent of the proportion of educational upgrading that is explained by observed task shifts.

5.3 A Note on Polarization

The question that has been neglected in the analysis so far is whether there has been a polarization of work in recent decades.³⁶ The argument is that the substitution of, for example, routine cognitive tasks by computer technologies affects occupations such as bookkeepers and bank employees that are traditionally held by employees with medium levels of education. Non-routine manual activities, on the other hand, that, at present, cannot be accomplished by computers, are frequent in occupations that are often held by employees with low levels of education such as waiters. In this respect, the Autor-Levy-Murnane task framework diverges from the prediction of the “traditional” skill-biased technological change hypothesis of an increasing demand for “skilled” jobs relative to “unskilled” jobs.

In order to investigate this question, I constructed a scalar index of occupational skill requirements. This skill index is the predicted value derived from the regression presented in Table B in the appendix (Panel A, for 1979). The only difference is that I additionally included the non-routine manual task category into the specification. This category has not been analyzed in detail previously because the task framework does not make predictions about the relationship between non-routine manual tasks and computer technologies. However, this category now has an important role for the question of “polarization”. As already stated, many occupations at the bottom of the skill distribution such as waiters or domestic staff involve non-routine manual tasks. So it is exactly the lag of a relationship between computer technologies and this task category that suggests the possibility of employment polarization.³⁷

Based on the 1979 value of this skill index, I classified occupations into ten groups. Occupations whose value of the skill index in 1979 was in the lowest decile are the first group, occupations in the second decile are the second group and so on.³⁸ This classification serves as the baseline occupational distribution, representing occupational skill requirements closest to the pre-computer era. Table 10 shows the distribution of task inputs in 1979 by deciles. The measure for non-routine manual task inputs is highest in the first

³⁶See, for example, Levy and Murnane (1992).

³⁷Goos and Manning (2003) provide, based on this reasoning, evidence for increased job polarization in the UK.

³⁸I also classified occupations according to mean hourly wages, a measure of the quality of occupations typically used in studies for the U.S. and UK. This alternative measure changes the classification of occupations particularly in the lowest deciles, whereas the allocation of occupations in the highest deciles changes only marginally. There are a number of occupations that rank in the lowest deciles based on the skill index but belong to deciles in the middle of the distribution based on average wages. Examples are ceramicists, blacksmiths and casters. Wages in these occupations are typically set in union pay agreements, which explains their high wage levels relative to skill requirements.

two deciles, while the measures for analytical and interactive task inputs are increasing (nearly monotonically) with deciles. Interestingly all deciles have relatively high values in routine task inputs, but routine manual activities are more frequent in lower deciles and routine cognitive activities are more frequent in upper deciles.

The question of whether there has been a “hollowing out” of middle-class occupations concerns the employment trends among these groups of occupations. Therefore, I calculated the distribution of employment among different groups of occupations over time. Figure 5 shows the employment changes in occupation groups between 1979 and 1998/99. This figure shows that employment has grown by nearly 3 percentage points in the 10th decile and by about 2.5 percent in the 9th decile of the skill index between 1979 and 1998/99. Examples of occupations grouped in these two categories are engineers, consultants, tax accountants, merchandisers, dealers and scientists. In addition, Figure 5 shows that employment has also grown in occupations belonging to the 1st decile of the skill index, although the rate (about 1 percentage point) was less pronounced than in the top two deciles of the skill distribution. Examples of occupations in the 1st decile are waiters, blacksmith, domestic staff, hoteliers and casters. By contrast, employment decline has been most pronounced in occupations in the 3rd decile of the skill distribution which includes occupations such as office clerks, machine operators and galvanizers. Thus, this tentative analysis suggests that the employment trends are consistent with a process of employment polarization.³⁹

³⁹An exception being the 7th decile; although ranking relatively high in the skill index distribution in 1979, occupations in this category have witnessed a decline in employment. However, having a closer look at occupations and task inputs in this group, this evidence is not surprising as cognitive and manual routine activities are frequent in this decile. Examples of occupations are bank clerks and insurance clerks.

6 Conclusions

West Germany (together with other continental European countries) has attracted attention because of the stability of its wage structure, raising the question of whether or not West Germany has experienced similar changes in skill demand to those in the U.S. and the U.K. in recent decades. This paper advances the debate by looking beyond wages and by offering a detailed characterization of changing occupational skill requirements in West Germany between 1979 and 1998/99. In addition, I provide detailed evidence on the nuanced version of the skill-biased technological change hypothesis, introduced in literature by Autor et al. (2003).

The results suggest that occupations today involve greater complexity than two decades ago. In recent decades, occupations have experienced a shift toward analytical and interactive activities, and away from cognitive and manual routine tasks. This development was ubiquitous in the sense that it occurred within occupations, within occupation-education groups and within occupation-age groups. In addition, the results indicate that the diffusion of computer technologies in the workplace has intensified these changes. This result comes from the fact that computers substitute for workers performing manual and cognitive routine tasks, but complement workers in performing analytical and interactive activities.

Owing to the direct skill measures, which are unlikely to be influenced by factors such as unions that typically distort the relationship between wages and skills, I am able to show that West Germany (and probably other continental European countries as well) has witnessed changes in skill requirements similar to those in the U.S. in recent decades. This evidence moves the debate over the differential wage developments in continental European countries and the U.S./U.K. forward to the question of why similar changes in skill requirements in all of these countries have not led to similar changes in the wage structure.

In addition, the result that occupations today require more skills highlights the importance of education and training to enable employees to cope with the challenges brought about by the changing task composition of occupations. It draws a particularly pessimistic picture for the labor market prospects of employees with low levels of education. This group of employees has experienced the least favorable labor market developments in recent decades. As the skill level of employees with low levels of education no longer meets the minimum occupational skill requirements, they become increasingly marginalized.

Table 1: Assignment of Activities

Classification	Tasks
non-routine analytic	researching, analyzing, evaluating and planning, making plans, constructions, designing, sketching working out rules/prescriptions using and interpreting rules
non-routine interactive	negotiating, lobbying, coordinating, organizing teaching or training selling, buying, advising customers, advertising entertaining or presenting employ or manage personnel
routine cognitive	calculating, bookkeeping correcting of texts/data measuring of length/weight/temperature
routine manual	operating or controlling machines equip machines
non-routine manual	repairing or renovation houses/apartments/machines/vehicles restoring of art/monuments serving or accomodating

Table 2: Descriptive Figures on Educational Trends and Computer Diffusion

Panel A: Proportion of Different Educational Groups in Employment				
	1979	1985/86	1991/92	1998/99
High Level of Education	8.18	8.85	13.30	16.48
Medium Level of Education	72.38	68.33	71.28	70.57
Low Level of Education	21.84	22.81	15.42	12.95
Panel B: Spread of Computers, Terminals, Laptops, Electronic Data-Processing Devices				
	1979	1985/86	1991/92	1998/99
Overall	6.06	18.11	34.52	55.38
High Level of Education	12.22	25.58	60.73	83.15
Medium Level of Education	6.31	20.00	33.77	56.52
Low Level of Education	3.44	10.19	16.13	32.65

The sample includes workers aged 18-65 with residence in West Germany and of German nationality.

Table 3: Trends in Aggregate Skill Inputs and by Education

	Analytic Skills	Interactive Skills	Routine Cognitive	Routine Manual	Non-routine Manual
Panel A: Overall					
1979	4.42	8.47	36.86	30.88	14.19
1985/86	9.71	10.47	31.81	26.18	19.90
1991/92	10.98	16.55	26.97	23.48	19.78
1998/99	13.93	33.81	22.11	17.19	26.04
Panel B: Employees with High Level of Education					
1979	15.45	20.10	42.51	15.59	4.10
1985/86	21.40	18.76	45.74	9.30	4.46
1991/92	26.29	35.23	34.33	8.70	5.75
1998/99	24.62	48.40	11.44	7.77	2.41
Panel C: Employees with Medium Level of Education					
1979	4.16	8.63	39.15	33.40	16.18
1985/86	9.29	10.41	33.31	28.85	24.18
1991/92	9.39	14.99	28.22	26.27	23.73
1998/99	11.88	28.34	24.08	19.44	26.80
Panel D: Employees with Low Level of Education					
1979	2.78	4.80	27.47	26.50	10.11
1985/86	6.29	7.34	21.58	24.74	13.27
1991/92	4.94	7.47	20.42	23.37	13.73
1998/99	6.92	14.44	14.74	18.19	18.77

The sample includes workers aged 18-65 with residence in West Germany and of German nationality.

Table 4: Trends in Aggregate Skill Inputs by Birth Cohorts

Analytical Task Inputs				
Year of Birth	1979	1985/86	1991/92	1998/99
1975-1981				7.41
1969-1974			7.71	11.54
1962-1968		5.58	10.16	14.63
1956-1961	2.39	9.72	13.35	15.09
1950-1955	5.15	11.16	12.69	13.95
1940-1949	5.49	11.23	11.22	15.32
before 1940	4.23	9.47	8.76	16.56
10 x Average Annualized Changes 1979-1998/99				
Within Cohorts:	5.90			
Within Age Levels:	4.89			
Interactive Task Inputs				
Year of Birth	1979	1985/86	1991/92	1998/99
1975-1981				23.17
1969-1974			9.52	30.35
1962-1968		6.25	15.05	33.61
1956-1961	5.72	10.13	18.36	34.37
1950-1955	8.77	11.96	20.14	36.68
1940-1949	9.90	11.95	17.81	37.26
before 1940	8.48	10.65	14.15	36.98
10 x Average Annualized Changes 1979-1998/99				
Within Cohorts:	18.33			
Within Age Levels:	16.61			
Routine Cognitive Task Inputs				
Year of Birth	1979	1985/86	1991/92	1998/99
1975-1981				22.44
1969-1974			22.21	21.41
1962-1968		25.06	28.86	24.01
1956-1961	38.04	33.08	29.04	23.54
1950-1955	42.08	35.72	29.69	20.57
1940-1949	39.87	33.99	27.73	19.22
before 1940	32.36	29.43	21.45	23.05
10 x Average Annualized Changes 1979-1998/99				
Within Cohorts:	-6.11			
Within Age Levels:	-7.01			

< Table continues on next page >

< Table 4 continued >

Routine Manual Task Inputs				
Year of Birth	1979	1985/86	1991/92	1998/99
1975-1981				18.97
1969-1974			29.80	17.23
1962-1968		31.51	25.23	19.15
1956-1961	41.53	27.21	24.99	17.68
1950-1955	35.13	26.05	21.80	15.19
1940-1949	30.52	23.91	20.87	14.71
before 1940	25.21	24.80	22.91	17.61
10 x Average Annualized Changes 1979-1998/99				
Within Cohorts:	-10.47			
Within Age Levels:	-8.39			
Non-Routine Manual Task Inputs				
Year of Birth	1979	1985/86	1991/92	1998/99
1975-1981				30.28
1969-1974			25.23	27.78
1962-1968		22.83	22.36	25.94
1956-1961	15.42	20.78	20.18	25.69
1950-1955	14.18	18.05	18.35	25.68
1940-1949	13.39	18.91	17.31	24.45
before 1940	14.25	19.74	19.37	23.30
10 x Average Annualized Changes 1979-1998/99				
Within Cohorts:	4.68			
Within Age Levels:	6.57			

The sample includes workers aged 18-65 with residence in West Germany and of German nationality.

Table 5: Shift-Share Analysis of Changes in Skill Requirements

10 x Annual Changes in Task Measures										
	analytic		interactive		routine cognitive		routine manual		non-routine manual	
	Overall									
1979-85	8.82		3.35		-8.43		-7.83		9.52	
1985-91	2.12		10.12		-8.05		-4.50		-0.21	
1991-99	4.21		24.63		-6.94		-8.98		8.94	
1979-99	5.01		13.34		-7.76		-7.20		6.23	
	Between and Within Occupational Decomposition									
	btwn	wthn	btwn	wthn	btwn	wthn	btwn	wthn	btwn	wthn
1979-85	-0.27	9.10	0.15	3.21	-1.40	-7.03	-1.26	-6.57	0.77	8.75
1985-91	0.44	1.68	0.10	10.02	0.87	-8.92	-0.00	-4.50	0.34	-0.55
1991-99	2.67	1.55	5.24	19.39	0.06	-7.00	-6.04	-2.94	-0.97	9.91
1979-99	0.77	4.24	1.70	11.64	-0.06	-7.70	-0.98	-6.22	0.12	6.11

The sample includes workers aged 18-65 with residence in West Germany and of German nationality. Occupations are defined according to the 2-digit level of the classification of occupational titles.

Table 6: Bivariate Regressions: Technological Change and Changes in Skill Requirements

Dep. Variables: (Annualized Changes in Task Inputs)				
	Δ analytic	Δ interactive	Δ routine cognitive	Δ routine manual
Panel A				
Δ computer use	0.086*** (0.032)	0.188*** (0.031)	-0.312*** (0.105)	-0.561*** (0.148)
dummy 1985/86-1991/92	-6.160*** (1.129)	3.536** (1.767)	-1.960 (3.098)	-2.462 (7.712)
dummy 1991/92-1998/99	-7.987*** (1.381)	8.915*** (1.440)	16.394** (7.726)	-7.436 (7.065)
R^2	0.183	0.337	0.079	0.131
Number of observations:	237			
Panel B				
Lagged Δ computer use	-0.022 (0.035)	0.160*** (0.031)	-0.796*** (0.202)	-0.173*** (0.065)
dummy 1985/86-1991/92	1.470 (1.350)	-4.632*** (1.703)	-23.444*** (7.237)	6.005* (3.637)
R^2	0.015	0.195	0.205	0.053
Number of observations:	156			
Weighted mean dependent variable	0.425	0.980	-0.849	-0.504

Robust standard errors are in parentheses; regressions are weighted by the number of individuals within occupation group; ***, **, *-indicate significance at the 1, 5, 10 percent level.

Table 7: Changes in Skill Requirements and Changes in the Educational and Gender Distribution

Dep. Variables: Annualized Changes in Task Inputs				
	Δ analytic	Δ interactive	Δ routine cognitive	Δ routine manual
Panel A				
Δ computer use	0.076*** (0.031)	0.181*** (0.033)	-0.311*** (0.108)	-0.529*** (0.163)
Δ Proportion w/ high educ. level	0.064** (0.033)	0.081** (0.039)	-0.195* (0.115)	0.049 (0.178)
Δ Proportion w/ medium educ. level	0.006 (0.025)	0.038 (0.034)	-0.177* (0.104)	0.257 (0.184)
Δ Proportion female employees	-0.119*** (0.049)	0.026 (0.049)	-0.607*** (0.191)	-0.068 (0.175)
R^2	0.222	0.348	0.112	0.148
Number of observations:	237			
Panel B				
Lagged Δ computer use	-0.009 (0.032)	0.153*** (0.031)	-0.752*** (0.211)	-0.187*** (0.072)
Lagged Δ Proportion w/ high educ. level	-0.056 (0.067)	0.011 (0.048)	-0.680*** (0.195)	0.244*** (0.094)
Lagged Δ Proportion w/ medium educ. level	-0.015 (0.036)	-0.103* (0.060)	-0.123 (0.140)	0.145** (0.066)
Lagged Δ Proportion female employees	0.110 (0.068)	0.177*** (0.054)	-0.714*** (0.268)	0.045 (0.178)
R^2	0.059	0.274	0.275	0.084
Number of observations:	156			

Control variables are: time dummies. Robust standard errors are in parentheses. Regressions are weighted by the number of individuals within occupation group. ***, **, *-indicate significance at the 1, 5, 10 percent level.

Table 8: Technological Change and Changes in Skill Requirements by Education

Dep. Variables: Annualized Changes in Task Inputs				
	Δ analytic	Δ interactive	Δ routine cognitive	Δ routine manual
Employees with High Level of Education				
Δ computer use	0.027 (0.065)	0.058 (0.060)	-0.253** (0.120)	-0.052 (0.077)
dummy 1985/86-1991/92	-0.356 (4.368)	17.304*** (4.483)	-10.983 (7.911)	0.393 (7.312)
dummy 1991/92-1998/99	-9.898*** (3.870)	10.594*** (3.987)	-17.912** (8.395)	4.761 (6.494)
R^2	0.084	0.171	0.084	0.010
Number of observations:	121			
Employees with Medium Level of Education				
Δ computer use	0.069* (0.040)	0.127*** (0.029)	-0.131 (0.114)	-0.521*** (0.171)
dummy 1985/86-1991/92	-7.465*** (1.159)	2.646* (1.628)	-2.880 (3.243)	-3.525 (8.226)
dummy 1991/92-1998/99	-8.208*** (1.400)	10.095*** (1.481)	18.047** (8.854)	-9.451 (8.006)
R^2	0.186	0.299	0.067	0.113
Number of observations:	234			
Employees with Low Level of Education				
Δ computer use	0.128*** (0.042)	0.151*** (0.044)	-0.196* (0.120)	-0.309*** (0.129)
dummy 1985/86-1991/92	-3.685*** (1.176)	0.230 (1.762)	1.498 (3.814)	-2.267 (7.916)
dummy 1991/92-1998/99	-5.472*** (1.438)	4.760*** (1.616)	31.453*** (7.780)	-11.904** (6.445)
R^2	0.149	0.148	0.161	0.056
Number of observations:	226			

Robust standard errors are in parentheses. Regressions are weighted by the number of individuals within occupation group. ***, **, * - indicate significance at the 1, 5, 10 percent level.

Table 9: Shifts in High-Educated-Equivalent Labor Demand and Medium-Educated-Equivalent Labor Demand Implied by Changes in Occupational Skill Requirement

	1979-1985/86	1985/86-1991/92	1991/92-1998/99	1979-1998/99
A. 10 x Observed Annual Changes in the Proportion of Employees with...				
High Level of Education	3.667	5.333	5.167	4.250
Medium Level of Education	-6.833	5.000	-1.167	-0.947
B. 10 x Observed Annual Changes in Within Occupation Skill Requirements				
analytic skills	9.037	1.716	1.774	4.242
interactive skills	3.046	9.981	16.430	10.705
routine cognitive	-6.059	-9.054	-10.416	-8.965
routine manual	-5.875	-5.216	-4.028	-7.305
C. Predicted Proportion of Changes in the Share of Employees with High (Medium) Level of Education Explained by Observed Changes in Within Occupation Skill Requirements (in Percent)				
High Level of Education	47.934	32.237	49.741	48.997
Medium Level of Education	8.815	-8.834	49.463	64.698
D. Predicted Annual Changes in Occupation Skill Requirements Implied by Computerization (10 x Annual Changes)				
analytic skills	8.901	2.991	0.519	4.129
interactive skills	4.243	9.572	15.013	9.664
routine cognitive	-5.547	-9.001	6.519	-2.375
routine manual	2.171	-0.134	-4.503	-0.964
E. Predicted Proportion of Changes in the Share of Employees with High/Medium Level of Education Explained by Predicted Changes in Occupation Skill Requirements Implied by Computerization (in Percent)				
High Level of Education	47.184	32.298	47.900	41.736
Medium Level of Education	6.669	-7.506	46.546	46.171

Table 10: Distribution of Task Inputs by Deciles

	Analytic Skills	Interactive Skills	Routine Cognitive	Routine Manual	Non-routine Manual
First decile	0.39	2.67	11.65	44.20	30.77
Second decile	0.55	2.72	8.93	30.12	26.55
Third decile	1.58	3.71	23.32	36.73	12.91
Fourth decile	1.92	3.78	20.57	26.10	17.07
Fifth decile	0.67	2.66	10.76	13.86	18.69
Sixth decile	0.88	2.30	14.20	14.04	8.16
Seventh decile	1.63	6.39	31.04	17.71	15.39
Eighth decile	5.65	9.10	37.08	22.33	7.84
Ninth decile	11.82	13.98	44.42	22.54	9.35
Tenth decile	17.73	26.92	42.11	12.03	2.82

Figure 1: Trends in Aggregate Skill Inputs

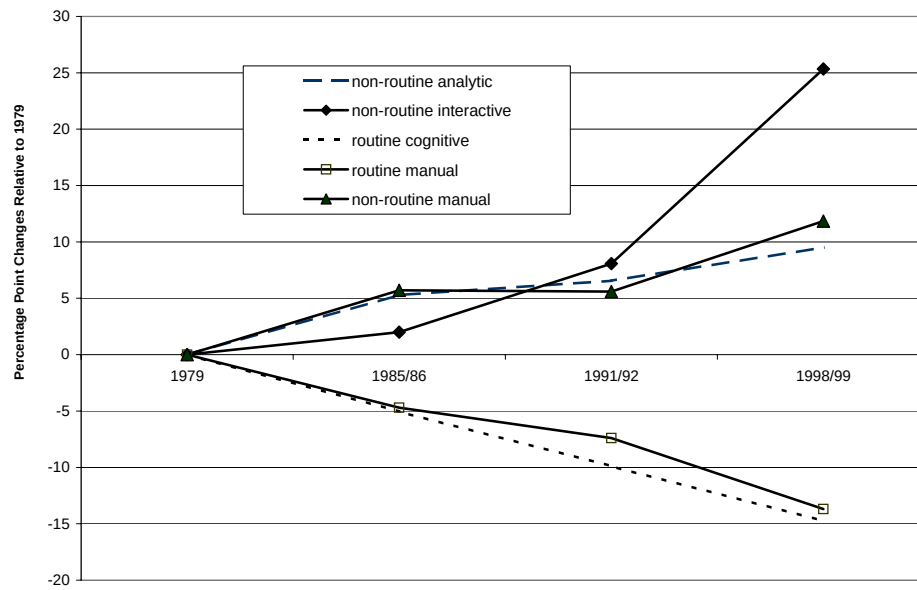


Figure 2: Employees with High Levels of Education: Trends in Aggregate Skill Inputs

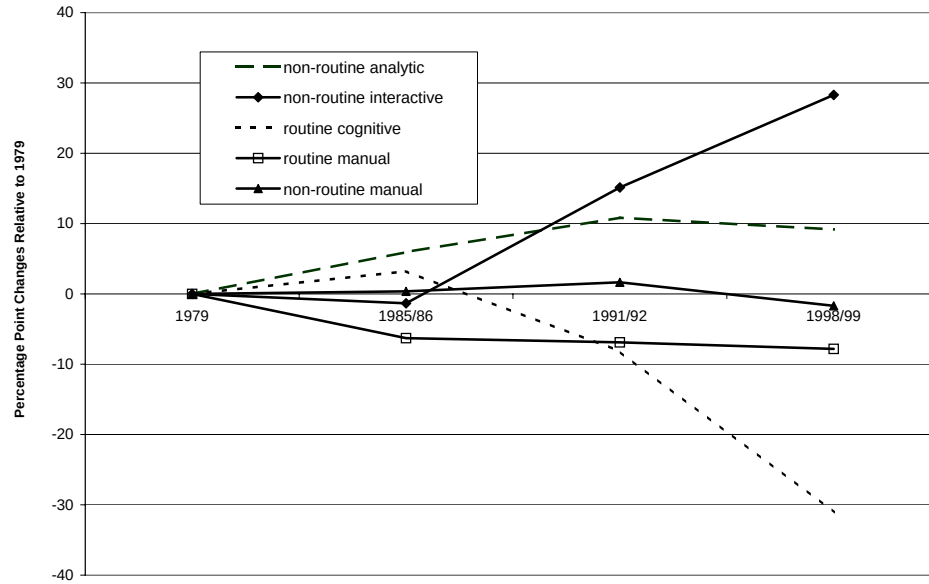


Figure 3: Employees with Medium Levels of Education: Trends in Aggregate Skill Inputs

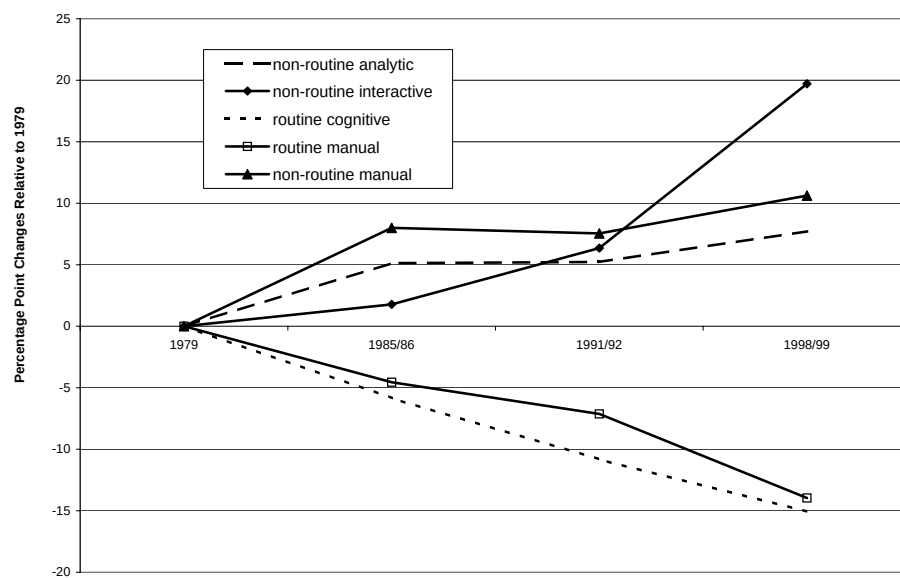


Figure 4: Employees with Low Levels of Education: Trends in Aggregate Skill Inputs

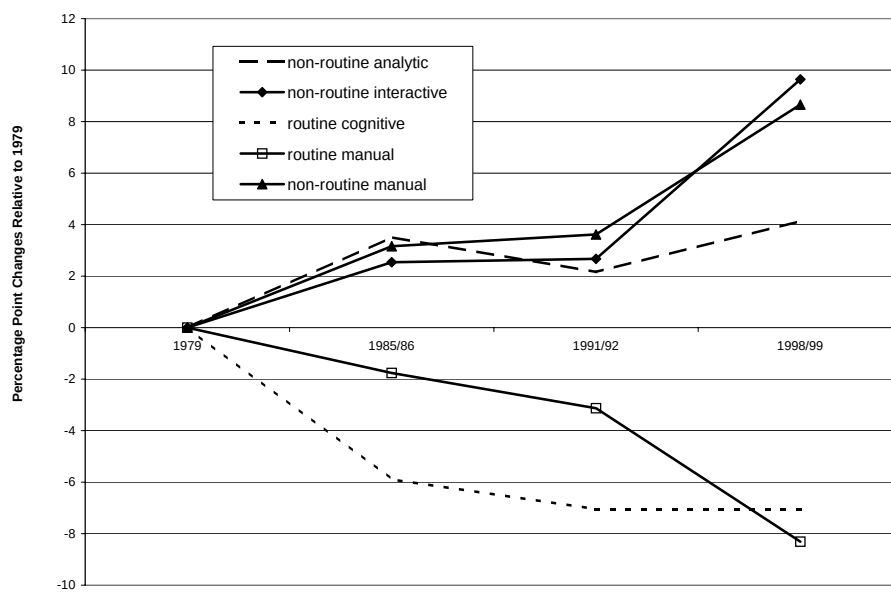
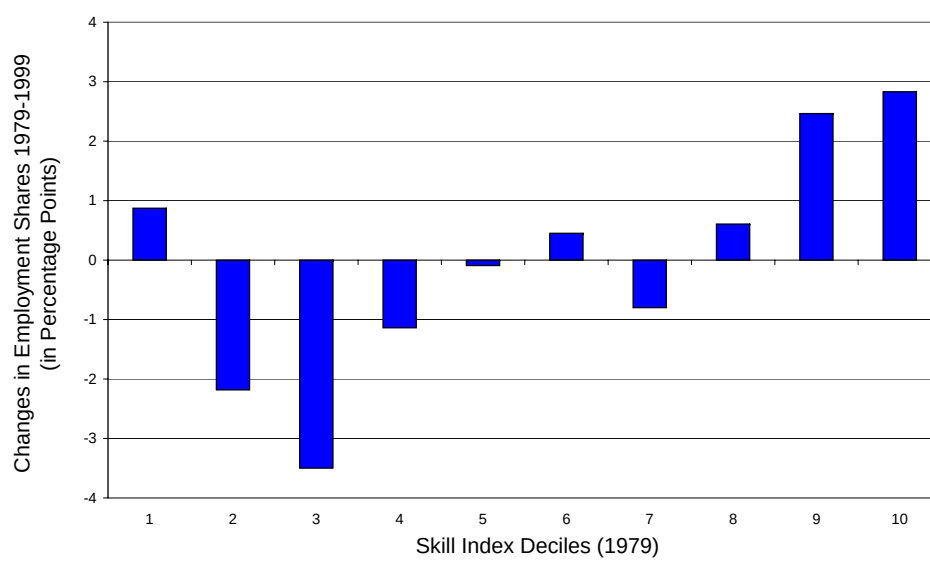


Figure 5: Changes in Employment Shares by Occupational Skill Index Deciles



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7 Appendix

Table A: Industry Classification

Manufacturing incl. construction and mining		Services	
10	Mining	29	Laundry and Dry Cleaning
11	Chemical Industry, Rubber and Synthetic Material	30	Hairdressing
12	Stone and Clay, Glass and Ceramics	40	Trade
13	Iron and Steel Production	52	Transport Services (including Carriage, Travel Agency, Warehouse)
14	Steel and Light Metal, Tracked Vehicles	53	Credit Institutions
15	Machine Construction	54	Insurance Companies
16	Car Industry	55	Catering and Hotels
17	Shipbuilding, Aircraft and Aerospace Industry	57	Health and Veterinary
18	Office and Data-Processing Machines	61	Radio, TV, Publishing House, Art, Theater, Museum
19	Electrical Engineering	62	Other Private Services
20	Precision and Optical Instruments		
21	Musical Instruments, Toys, and Jewellery		
22	Construction		
23	Wood Processing		
24	Cellulose and Paper Industry		
25	Printing		
26	Leather and Shoe Industry		
27	Textile Industry		
28	Food, Beverages and Tobacco		
		Public and Quasi-Public Institutions	
		50	Postal Services
		51	Railway Services

Table B: OLS Results: Educational Requirements within Occupation Group as a Function of Task Inputs

A. Dep. Variable: Proportion of Employees with High Level of Education				
	1979	1985/86	1991/92	1998/99
analytic skills	1.321*** (0.337)	1.639*** (0.416)	1.102*** (0.217)	1.267*** (0.279)
interactive skills	1.322*** (0.307)	0.129 (0.566)	1.464*** (0.232)	0.706*** (0.186)
routine cognitive	-0.061 (0.110)	-0.301* (0.159)	-0.576*** (0.131)	-0.039 (0.223)
routine manual	-0.206 (0.120)	-0.476*** (0.157)	-0.021 (0.077)	0.086 (0.232)
R^2	0.468	0.388	0.673	0.583
Number of observations:	83	83	80	80

B. Dep. Variable: Proportion of Employees with Medium Level of Education				
	1979	1985/86	1991/92	1998/99
analytic skills	-0.436 (0.466)	-1.007* (0.534)	-0.622* (0.328)	-1.110*** (0.364)
interactive skills	-0.241 (0.424)	-0.047 (0.726)	-0.980*** (0.351)	-0.087 (0.343)
routine cognitive	0.019 (0.152)	0.631*** (0.205)	0.642*** (0.198)	0.290 (0.291)
routine manual	0.209 (0.167)	1.013*** (0.202)	0.171 (0.116)	-0.194 (0.303)
R^2	0.065	0.272	0.251	0.249
Number of observations:	83	83	80	80

Robust Standard Errors are in Parentheses; Each regression includes an intercept term. ***, ** indicate significance at the 1, 5 percent level.